TERM 4

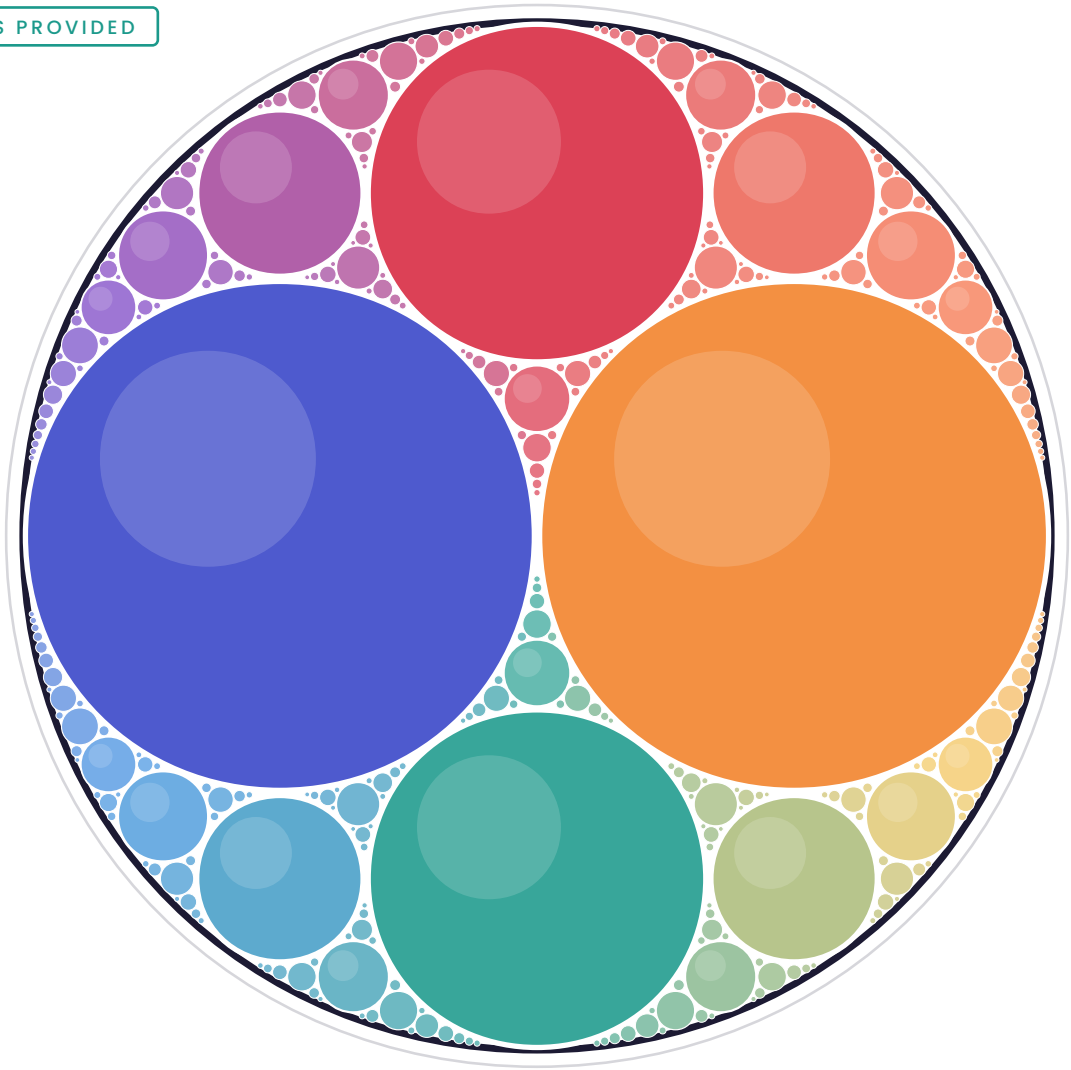
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 8

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

APOLLONIAN GASKET
every circle touches three others; every curvature is a whole number

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

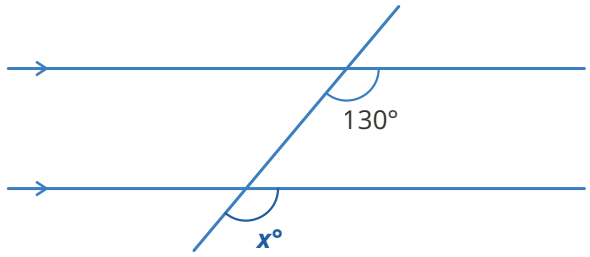
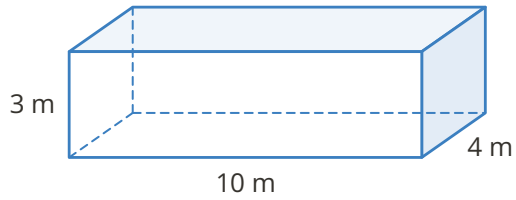
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Corresponding angles are equal COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism.  	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $-5 \times 4 + 6 \times 3$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Share 84 lollies in the ratio 4 : 3. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 3x + 2$. | x | 0 | 1 | 2 | 3 | | y | 2 | 5 | | |	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

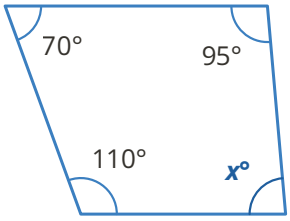
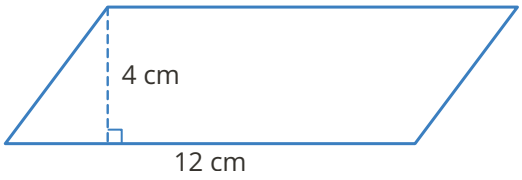
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 10% of $350	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Expand. $8(m + n)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mode of these scores. 5 8 3 8 6 2 8	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

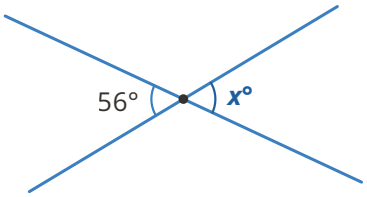
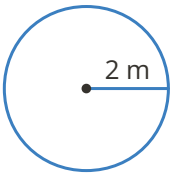
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 $\frac{3}{4}$ of the 32 nippers at the surf club passed the swim test. How many passed?	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 4 L of juice cost \$12. Find the price per litre.	FORMULA Unit rate = amount ÷ number of units COPY SUBSTITUTE ANSWER
5 Find the probability that the team does not win. $P(\text{win}) = \frac{7}{10}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

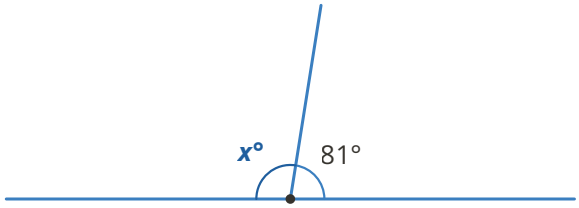
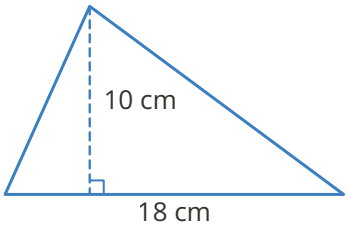
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. 3^4	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Solve. $4x - 7 = 13$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the mean of these goal tallies. Goals: 7, 9, 4, 10, 5	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

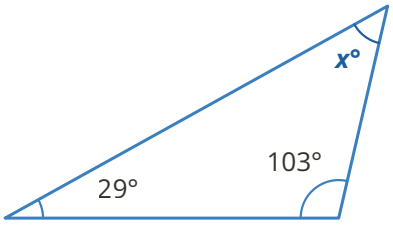
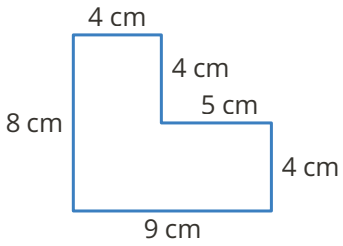
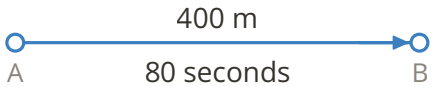
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Write as a decimal. $\frac{9}{25}$	FORMULA Fraction to decimal: numerator ÷ denominator COPY SUBSTITUTE ANSWER
4 A runner runs from A to B. Find the average speed. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 A six came up 15 times in 60 rolls of a die. Find the experimental probability of rolling a six.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

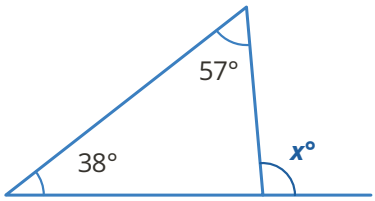
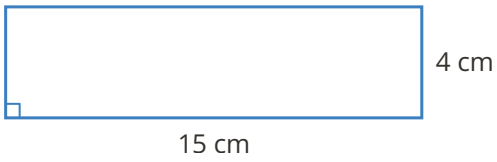
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this rectangle.  	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. 7 out of 35	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Find the value of this expression when n = 4. $\frac{12}{n} + 3$	FORMULA $\frac{12}{n} + 3$ COPY SUBSTITUTE ANSWER
5 Find the range of these test marks. Marks: 58, 72, 65, 49, 70	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

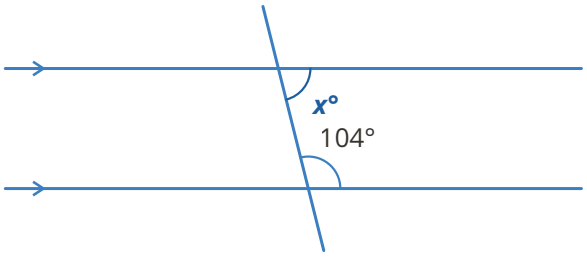
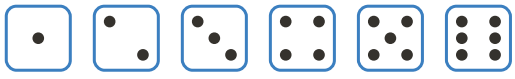
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Convert 1.25 m to centimetres.	FORMULA 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1000 m COPY SUBSTITUTE ANSWER
3 Calculate. 15% of $80	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 A touch footy carnival runs from 10:40 to 12:15. How long is that?	FORMULA Time taken = finish time − start time (60 minutes = 1 hour) COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a number less than 3. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

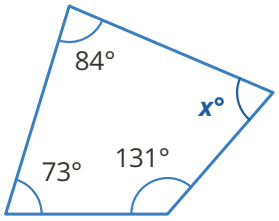
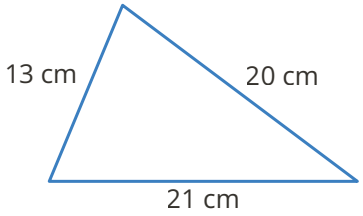
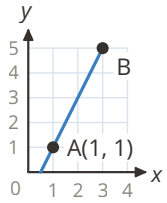
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the perimeter of this triangle.  	FORMULA Perimeter = add the lengths of all the sides COPY SUBSTITUTE ANSWER
3 Calculate. $56.2 \div 10$	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Expand. $6(a + 1)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = 2x - 1$. Write the coordinates of B.  	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

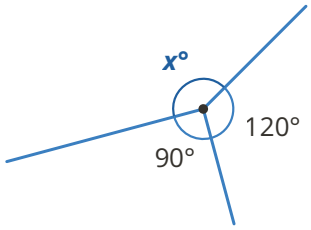
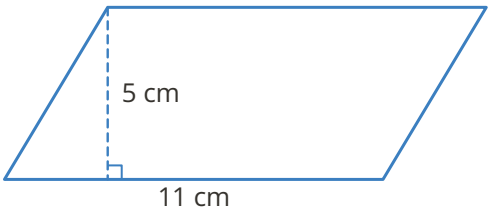
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram.  	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $(5 - 9) \times 3$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Simplify. $15 : 25$	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 Find the probability that the surf is not flat. $P(\text{flat}) = \frac{3}{8}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

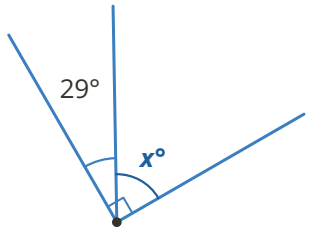
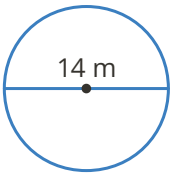
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{9}{10}$ of 70 kg	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Solve. $x - 4 = 9$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the median of these scores. Scores: 21, 25, 19, 30, 24	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

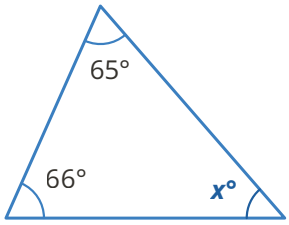
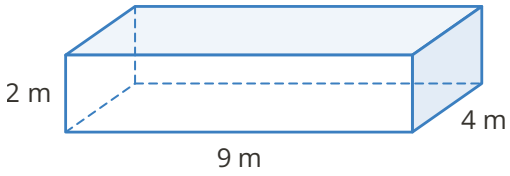
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism.  	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Write as a decimal. 64%	FORMULA Percentage to decimal: percentage ÷ 100 COPY SUBSTITUTE ANSWER
4 Share \$150 in the ratio 2 : 3. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Scores: 13, 19, 16, 10, 17	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

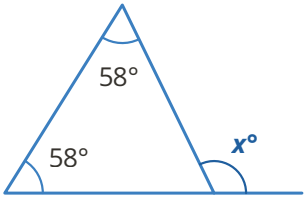
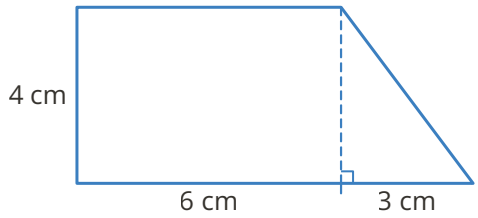
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 A netball shooter scored 21 goals from 25 shots. What percentage of the shots were goals?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Find the value of this expression when a = 6. $a^2 + 5$	FORMULA $a^2 + 5$ COPY SUBSTITUTE ANSWER
5 Find the range of these netball scores. Goals: 26, 18, 31, 22, 24	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

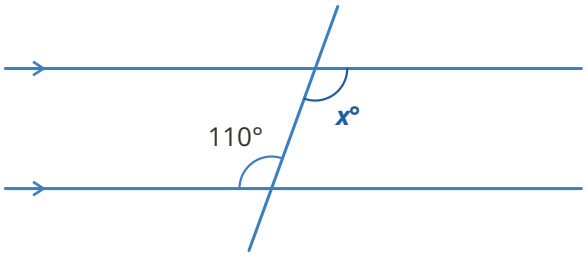
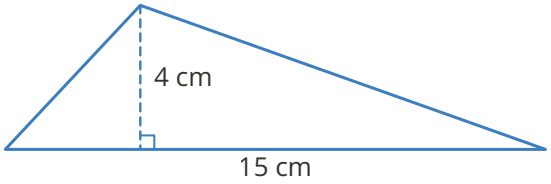
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Alternate angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this triangle.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 12% of \$50	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Which bag of potatoes is the better buy? Bag A: 3 kg for \$12 Bag B: 5 kg for \$15	FORMULA Price per unit = price ÷ number of units; best buy = lowest price per unit COPY SUBSTITUTE ANSWER
5 Heads came up 42 times in 100 tosses of a coin. Find the experimental probability of heads.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

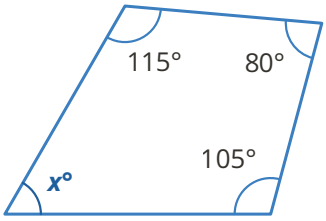
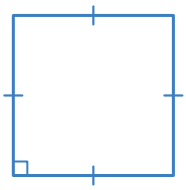
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this square.  	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Evaluate. $\sqrt[3]{64}$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Expand. $5(3x - 2)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is green. Colour: Red 4, Blue 2, Green 6	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

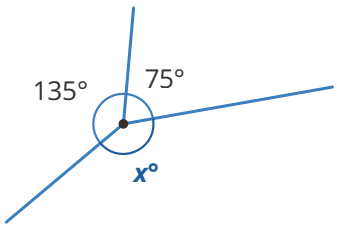
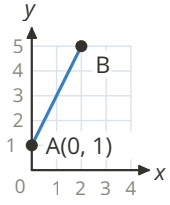
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the perimeter of this rectangle.  	FORMULA $P = 2(l + w)$ COPY SUBSTITUTE ANSWER
3 At the Kingscliff markets, $\frac{3}{5}$ of the 40 stalls sell food. How many stalls sell food?	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Solve. $6y + 4 = 40$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = 2x + 1$. Write the coordinates of B.  	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

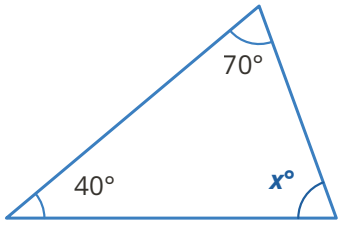
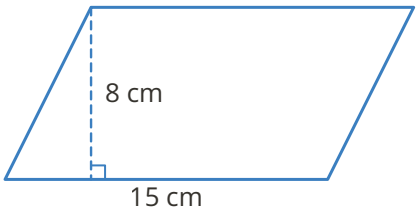
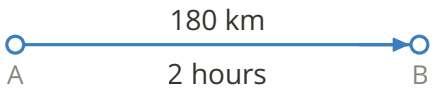
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram.  	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. $\frac{7}{20}$	FORMULA Fraction to percentage: fraction × 100% COPY SUBSTITUTE ANSWER
4 A car travels from A to B. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the probability that the counter is not red. $P(\text{red}) = \frac{1}{4}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

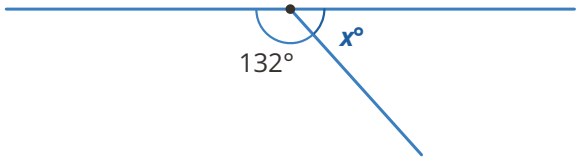
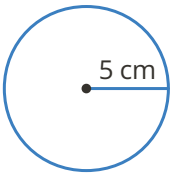
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Evaluate. $50 - 4 \times (7 + 3)$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $a = 5$, $b = 4$ and $c = 7$. $2(a + b) - c$	FORMULA $2(a + b) - c$ COPY SUBSTITUTE ANSWER
5 Find the mode of these scores. 2 4 4 1 3 2 4 5	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

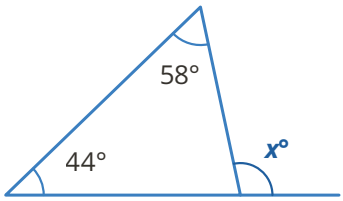
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Convert 2.5 L to millilitres.	FORMULA $1\ cm^3 = 1\ mL,\ 1\ L = 1000\ mL$ COPY SUBSTITUTE ANSWER
3 The Kingscliff touch footy team won 9 of its 15 games. What percentage of its games did it win?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Write 20:50 in 12-hour time (am or pm).	FORMULA 24-hour time: pm hours add 12 (1:00 pm = 13:00); 12:00 am = 00:00 COPY SUBSTITUTE ANSWER
5 Find the range of these maximum temperatures. °C: 19, 25, 22, 31, 27	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

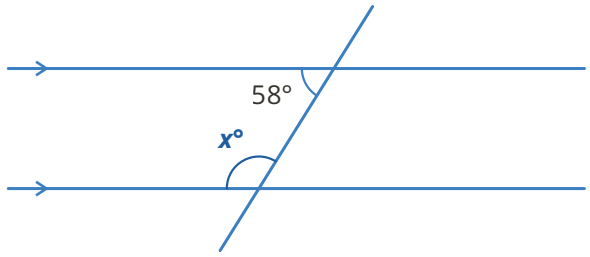
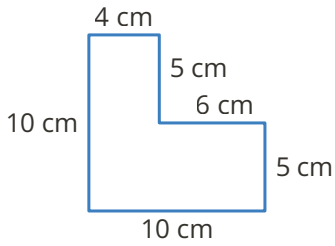
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. 0.25×1000	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Simplify. $32 : 12$	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 A spinner was spun 50 times. Find the experimental probability of blue. Colour: Red 12, Blue 18, Green 20	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

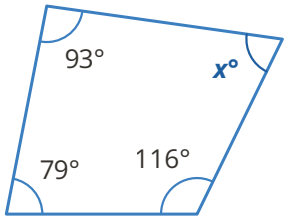
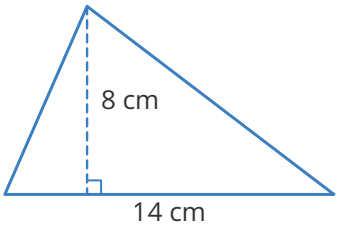
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this triangle.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 40% of \$75	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Expand. $3(x + 5)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mean of these try tallies. Tries: 2, 0, 4, 1, 3	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

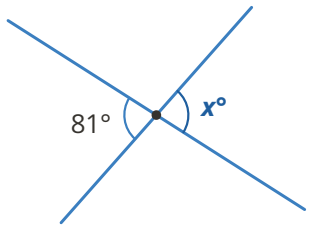
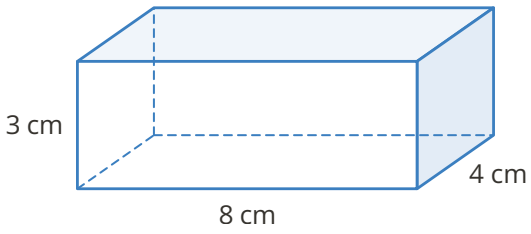
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{3}{7}$ of 56	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Share \$240 in the ratio 5 : 3. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 4x$. x: 1, 2, 3, 4 y: 4, , , 16	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

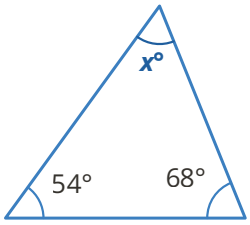
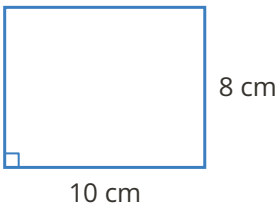
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this rectangle. 	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 0.48	FORMULA Decimal to fraction: write over 10, 100 or 1000, then simplify COPY SUBSTITUTE ANSWER
4 Solve. $y - 15 = 8$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A letter is chosen at random from this word. Find $P(\text{M})$. MATHEMATICS	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

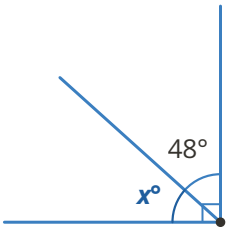
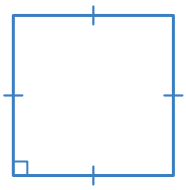
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the perimeter of this square.  	FORMULA $P = 4s$ COPY SUBSTITUTE ANSWER
3 Evaluate. $5^2 + 2^4$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 A car travels 360 km on 30 L of fuel. Find the rate in kilometres per litre.	FORMULA Unit rate = amount ÷ number of units COPY SUBSTITUTE ANSWER
5 Find the probability that the bus is not late. $P(\text{late}) = \frac{3}{20}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

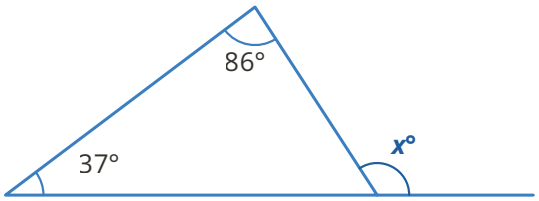
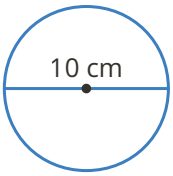
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 A rugby league team kicked 34 of its 40 goal attempts. What percentage did it kick?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Find the value of this expression when a = 6 and b = 3. $ab - 4$	FORMULA $ab - 4$ COPY SUBSTITUTE ANSWER
5 Find the range of these scores. Scores: 40, 38, 45, 33, 41	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

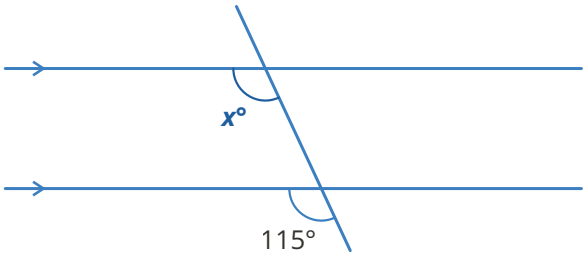
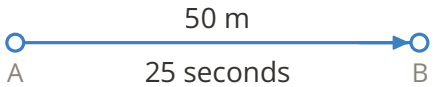
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Corresponding angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram.  	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $15 - 3 \times (2 + 4)$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 A swimmer swims from A to B. Find the average speed.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the median of these scores. 12, 15, 11, 18, 14, 20, 16	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

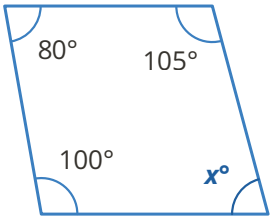
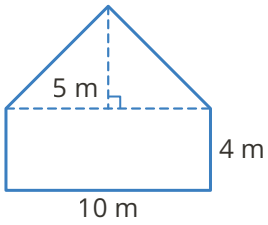
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. 90% of 50 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Expand. $7(2 + k)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mean of these netball scores. Points: 12, 15, 9, 18, 6	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

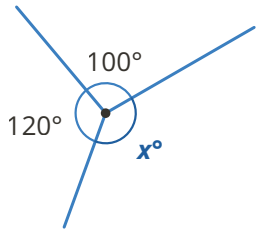
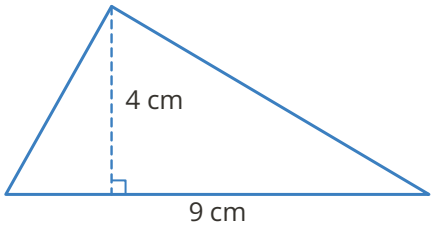
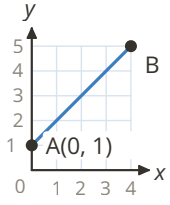
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the area of this triangle.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{3}{4}$ of 60	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Solve. $2x + 9 = 25$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = x + 1$. Write the coordinates of B.  	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

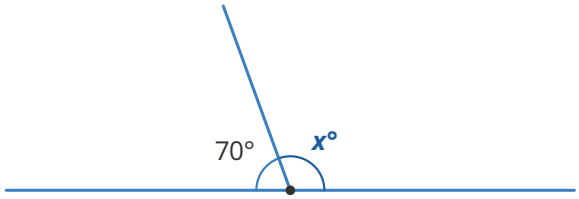
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Convert 3.5 m to centimetres.	FORMULA 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1000 m COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 15%	FORMULA Percentage to fraction: write over 100, then simplify COPY SUBSTITUTE ANSWER
4 Surf club training starts at 06:30 and finishes at 07:05. How long does it last?	FORMULA Time taken = finish time − start time (60 minutes = 1 hour) COPY SUBSTITUTE ANSWER
5 A die was rolled 60 times. Find the experimental probability of rolling a 3. Number: 1, 2, 3, 4, 5, 6 Frequency: 7, 9, 12, 10, 8, 14	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

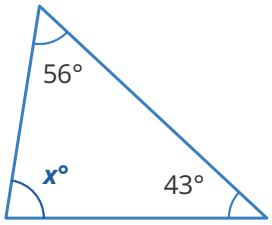
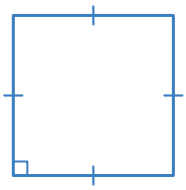
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this square.  	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Calculate. 5.19×1000	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Simplify. $8 : 20$	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is blue. Colour: Blue, Green, Red Marbles: 6, 4, 2	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

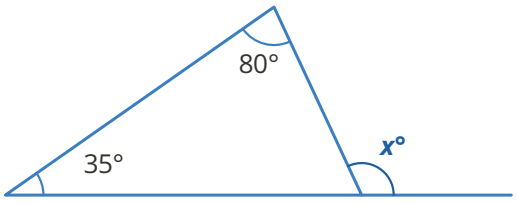
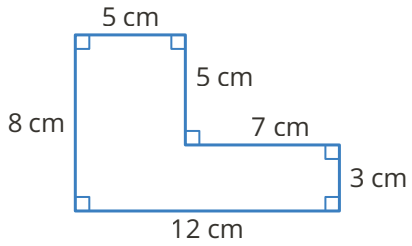
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the perimeter of this shape. 	FORMULA Perimeter = add the lengths of all the sides COPY SUBSTITUTE ANSWER
3 Write as a percentage. 19 out of 20	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Find the value of this expression when k = 4. $20 - 3k$	FORMULA $20 - 3k$ COPY SUBSTITUTE ANSWER
5 Find the range of these heights. cm: 104, 96, 118, 101, 99	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER